

Momentum ledger at the measured operating point
(0.217 mA / 800 V, tail-averaged)

$F_{\text{dep}} = +2.91 \text{ nN}$
(ambient e^- /ion impact)

$F_{\text{emit}} = +0.40 \text{ nN}$
craft
(cold gun + aperture)

$F_{\text{out}} = +17.53 \text{ nN}$
(escaped beam, all species)

control volume

$F_{\text{em}}(\text{craft}) = -20.0 \text{ nN}$
(field pull, from $\rho E_z + \text{stress}$)

— particle momentum flux (measured)

- - - field force (computed)

closed thrust = 17.13 nN $= F_{\text{beam}} - 2.3\%$

